

Testing a Structural DeFi Lending Model: Reduced-Form Evidence and Structural Limitations

Mitchell Ballif

December 10, 2025

Abstract

This paper examines whether a structural model of decentralized finance (DeFi) lending can explain observed stablecoin interest rate dynamics. Using daily data from Aave, I first document reduced-form links between U.S. risk-free rates and DeFi lending rates. I then evaluate the static equilibrium model of Barbon et al. (2025) using the Simulated Method of Moments. The reduced-form evidence shows a consistent relationship between monetary policy rates and on-chain lending conditions. However, the structural model cannot replicate this pattern or other basic features of the data, and key parameters are weakly identified. These failures point to missing mechanisms, particularly dynamics, heterogeneous liquidity providers, and frictions. The results suggest that richer, explicitly dynamic models are needed before DeFi lending can be used for credible structural interpretation or policy analysis.

1 Introduction

Decentralized finance (DeFi) lending protocols now intermediate tens of billions of dollars in U.S.-dollar-pegged “stablecoins.” Stablecoins are digital tokens designed to trade one-for-one with the U.S. dollar. In the markets I study, this peg is maintained by either (i) backing the token with off-chain dollar assets such as bank deposits and Treasury bills (USDC, USDT) or (ii) over-collateralizing the token with on-chain crypto assets (DAI). In practice, large traders treat stablecoins as a close substitute for cash or money market fund shares that can move natively on blockchains. DeFi lending protocols such as Aave transform this stablecoin liquidity into interest-bearing loans: liquidity providers (LPs) deposit stablecoins into a pool, borrowers post crypto collateral, and an algorithm sets the lending rate as a function of the pool’s utilization.

These markets raise a natural question for monetary policy; what actually drives lending rates on these platforms: arbitrage by liquidity providers, or shifts in borrower demand for leverage? This paper investigates that broader mechanism by asking a more focused research question: does the Barbon et al. (2025) model of interest rate determination match the data well enough to explain how stablecoin lending rates move with U.S. risk-free rates? If the model fits, it provides a tractable framework for analyzing on-chain money markets. If it does not, its failures reveal which economic forces and frictions must be incorporated to capture DeFi rate dynamics more accurately.

This paper specifically examines the static equilibrium model of Barbon, Barthélemy, and Nguyen (2025). My analysis uses daily panel data on USDC, USDT, and DAI lending rates from Aave and implement two complementary empirical strategies. First, I estimate reduced-form panel regressions that document how the Secured Overnight Financing Rate (SOFR) passes through to DeFi deposit rates. This provides the key empirical moments that any structural model must match. Second, I estimate the structural parameters of the Barbon et al. model using the Simulated Method of Moments (SMM).

Barbon et al. (2025) model risk-averse LPs who supply stablecoins to a lending pool and

investors who borrow against crypto collateral to fund leveraged positions. The equilibrium utilization rate $u = B/S$ determines interest rates through the protocol’s algorithmic rate function $\rho = u \cdot f(u) \cdot (1 - r)$, where r is the reserve factor. On the supply side when the risk-free rate ρ^* rises, LPs require higher compensation to keep funds in DeFi rather than reallocating to Treasuries, so equilibrium DeFi rates increase. On the demand side higher risk-free rates can reduce expected returns on crypto leverage, lowering borrowing demand and utilization and thereby decreasing DeFi rates. The sign of passthrough depends on which channel dominates.

My findings are as follows. The reduced-form regressions confirm positive passthrough: a 100 basis point increase in SOFR is associated with a 19 basis point increase in DeFi deposit rates. When I allow for rate persistence by including the lagged dependent variable, the implied long-run passthrough rises to approximately 0.68, though this estimate is imprecise. These magnitudes are broadly consistent with the comparative statics emphasized by Barbon et al. (2025) and indicate substantial, though incomplete, arbitrage between DeFi and conventional money markets.

The structural estimation, by contrast, fails to rationalize these patterns. The SMM procedure converges, but the estimated model cannot match three of the four moments. Most strikingly, the model generates a near-zero or slightly negative passthrough coefficient when the data show robust positive passthrough. The model also severely underpredicts rate volatility and overpredicts mean spreads. Moreover, key structural parameters are weakly identified: risk aversion and the liquidation risk premium have standard errors that are an order of magnitude larger than their point estimates, even after adjusting the covariance matrix for simulation noise.

These failures are informative. They reveal that the static equilibrium mechanism, coupled with the assumption of instantaneous LP rebalancing between DeFi and Treasuries, is not a good description of short-run DeFi rate dynamics. Instead, the evidence points to a market in which utilization is moved primarily by demand shocks from crypto speculation,

supply adjusts sluggishly, and frictions such as gas costs and regulatory uncertainty slow arbitrage. A structural model that captures these forces must go beyond the static model to incorporate dynamics and heterogeneous LPs.

This paper contributes to two literatures. First, it adds to the growing body of work on monetary policy transmission to crypto markets. Karau (2021) and Benigno and Rosa (2023) find that Bitcoin prices are largely orthogonal to monetary policy shocks, while Aldasoro et al. (2025) show that contractionary policy correlates with stablecoin outflows. I complement this work by studying interest rate passthrough to on-chain money markets, where arbitrage forces should create tighter links to conventional rates than exist for speculative crypto assets. Second, the paper contributes to the structural estimation literature by using SMM as a diagnostic tool for model misspecification. Following Reiss and Wolak (2007), I interpret gaps between data and model moments as evidence about which economic mechanisms are missing.

The remainder of the paper proceeds as follows. Section 2 describes the data. Section 3 lays out the two estimation strategies, their identifying assumptions, and key limitations. Section 4 reports results and interprets the model’s failures. Section 5 summarizes the main limitations. Section 6 concludes.

2 Data

I construct a daily panel of stablecoin lending rates and related variables from June 2021 to May 2025. The underlying DeFi data comes from hourly observations retrieved through the Aave Subgraph API for Aave v2 and Aave v3, the largest lending protocols on Ethereum. Because DeFi markets can have substantial intraday volatility driven by liquidations and gas-price spikes, I aggregate the hourly data to the daily level. This smoothing reduces noise, and aligns the on-chain data with the lower-frequency macroeconomic variables used in my analysis.

The dataset covers three major USD-pegged stablecoins: USDC, USDT, and DAI. For each stablecoin and protocol, I observe the deposit rate earned by liquidity providers, the utilization rate defined as total borrows divided by total liquidity, and the total value locked (TVL). Macroeconomic series, including the Secured Overnight Financing Rate (SOFR) and the 5-year breakeven inflation rate, are obtained from the Federal Reserve Economic Data (FRED) database. Crypto-market controls include Bitcoin prices, Bitcoin’s 30-day realized volatility, and Ethereum gas prices (in gwei), which capture network congestion and transaction costs. I also use the VIX index to measure broader risk sentiment that may spill over into digital-asset markets.

After merging these sources and removing observations with missing data, the final panel contains approximately 4,900 stablecoin-day observations. I further construct a measure of liquidation risk, which captures the protocol’s exposure to sudden deleveraging episodes. This measure plays a central role in the structural model because the liquidation risk premium $\theta\psi$ determines how utilization risk affects liquidity providers’ willingness to supply capital.

Summary statistics appear in Table 1. DeFi deposit rates average 3.9 percent annually with a standard deviation of 3.1 percentage points. Utilization rates average roughly 75 percent, implying that most available liquidity is actively borrowed. SOFR varies substantially over the sample, rising sharply during the 2022 - 2023 tightening cycle. Gas prices, Bitcoin volatility, and the VIX all display high variance, reflecting the highly dynamic environment of DeFi lending markets.

Table 1: Summary Statistics

Variable	N	Mean	Std. Dev.	Min	Max
DeFi deposit rate (%)	4,944	3.92	3.05	0.15	17.88
SOFR (%)	3,360	3.49	2.02	0.01	5.40
Utilization rate	4,944	0.75	0.14	0.20	0.96
Log TVL	4,944	22.00	0.72	20.39	23.40
Liquidation risk	4,827	0.51	0.16	0.17	0.99
Gas price (gwei)	4,923	27.73	32.47	0.07	207.00
BTC volatility (%)	4,827	51.47	16.45	16.98	99.13
VIX	3,462	19.21	5.40	11.86	52.33

Notes: DeFi data are aggregated from hourly observations pulled via the Aave Subgraph API for Aave v2 and v3. Macroeconomic and financial series are sourced from FRED. All variables are constructed at the stablecoin-day level.

Figure 1 plots the daily DeFi deposit rate against U.S. short-term rates. Three features emerge. First, DeFi rates moved largely independently of near-zero policy rates through mid-2022. Second, during the 2022 - 2023 tightening cycle, DeFi rates became increasingly correlated with SOFR, consistent with strengthening arbitrage linkages. Third, in 2024 and 2025, DeFi rates frequently traded below SOFR, with sharp spikes driven by episodes of intense leverage demand on-chain.

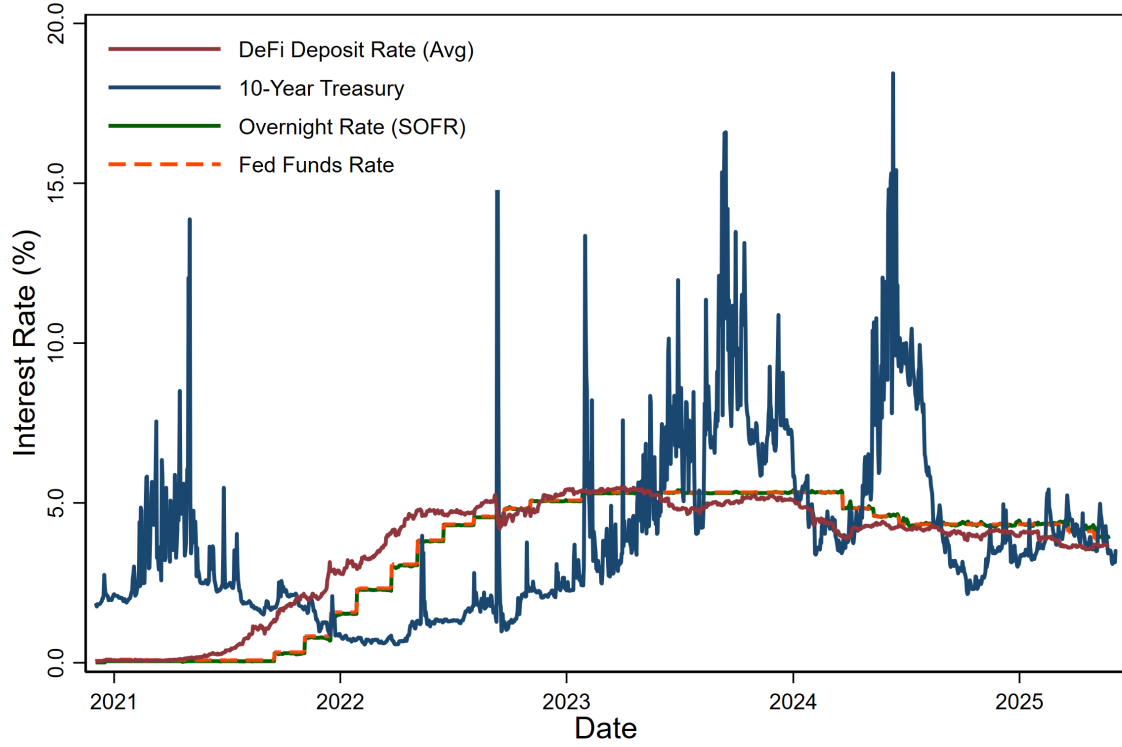


Figure 1: DeFi Deposit Rates and U.S. Short-Term Rates

Notes: The figure displays the daily average deposit rate across USDC, USDT, and DAI on Aave alongside the Federal Funds Rate and SOFR.

Figure 2 shows utilization rates for each stablecoin. Utilization varies substantially across coins and over time, reflecting changes in borrowing demand, collateral conditions, and liquidity availability. This variation is central to the structural model, as the algorithmic rate rule $\rho = u \cdot f(u) \cdot (1 - r)$ links interest rates directly to utilization.

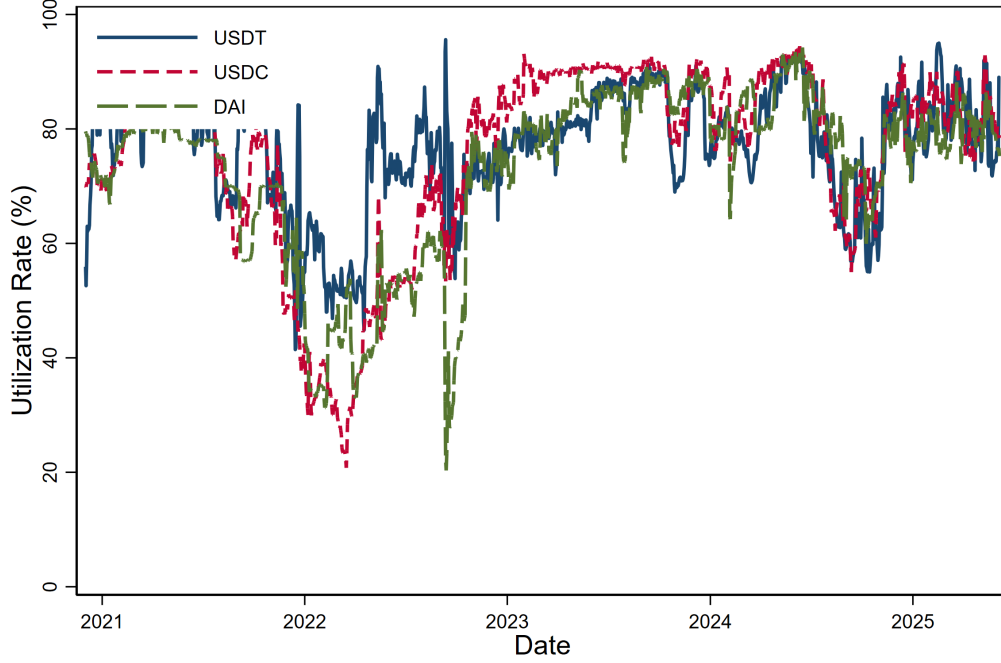


Figure 2: Utilization Rates by Stablecoin

Notes: Utilization is defined as total borrows divided by total liquidity in the lending pool.

3 Research Design

I employ two estimation strategies. The first, panel OLS with fixed effects, delivers estimates of policy passthrough. The second, Simulated Method of Moments, tests whether the Barbon et al. (2025) structural model can match key features of the data.

3.1 Strategy 1: Reduced-Form Panel Regression

The baseline estimating equation is:

$$\rho_{ct} = \alpha_c + \beta_1 \cdot \text{SOFR}_t + X_{ct}\Gamma + \varepsilon_{ct}, \quad (1)$$

where ρ_{ct} is the DeFi deposit rate for stablecoin c on day t , α_c is a coin fixed effect, and X_{ct} includes BTC volatility, BTC price, liquidation risk, gas prices, VIX, and inflation

expectations. The coefficient β_1 measures the passthrough of the risk-free rate to DeFi rates.

I also estimate a dynamic specification that includes the lagged dependent variable:

$$\rho_{ct} = \alpha_c + \beta_1 \cdot \text{SOFR}_t + \phi \cdot \rho_{c,t-1} + X_{ct}\Gamma + \varepsilon_{ct}. \quad (2)$$

Under a stable AR(1) approximation, the implied long-run passthrough in this specification is $\beta_1/(1 - \phi)$.

3.1.1 Identification and Limitations

The coefficient β_1 is informative about monetary policy passthrough and useful to the model even without rigorous conditional independence evidence for two key reasons: 1) the Federal Reserve views algorithmic stable coins as "endogenous to the same system" (Watsky et al. 2024) and do not provide a signal to the broader economy. Monetary Policy is therefore exogenously decided based on broader macro economic trends. 2) Confounding effects can be reasonably be controlled by money market, equity, and defi price information. It is therefore not unreasonable that, conditional on the covariates X_{ct} and coin fixed effects,

$$E[\epsilon_i \mid X_{ct}, \alpha_c] = 0$$

The control vector X_{ct} absorbs crypto-specific volatility, broad risk sentiment, and inflation expectations that might co-move with both SOFR and DeFi rates. However I do not make nor rely on this assumption to validate my structural model with the reduced form estimates

In practice, identification comes from time-series variation in SOFR that is common to all stablecoins, combined with cross-coin differences in how DeFi rates load on that common factor. I include coin fixed effects to control for time-invariant differences in mean rates across USDC, USDT, and DAI. I intentionally do not include time fixed effects, since they would soak up most of the variation in SOFR. This is a limitation: any omitted time-varying

factor that moves both SOFR and DeFi rates in the same direction and is not captured by X_{ct} will bias $\hat{\beta}_1$ upward. The dynamic specification in equation (2) adds the lagged dependent variable to capture persistence in DeFi rates. Under the AR(1) approximation, the implied long-run passthrough is $\beta_1/(1 - \phi)$.

Standard errors are clustered at the stablecoin level to allow for arbitrary serial correlation and heteroskedasticity within each coin. With only three clusters, conventional cluster-robust inference may understate uncertainty (Angrist and Pischke, 2009).

3.2 Strategy 2: Structural Estimation via SMM

3.2.1 Model

The Barbon et al. (2025) model features risk-averse LPs who supply stablecoins and investors who borrow to fund leveraged positions. LPs have mean-variance preferences, so their supply depends on the risk-free rate ρ^* , the liquidation wedge $\theta\psi$, and risk aversion γ :

$$S = \frac{\rho - \rho^* - \theta\psi}{\theta(1 - \theta)\gamma\psi^2},$$

where ρ is the DeFi deposit rate and $\theta\psi$ captures expected losses from liquidations. Borrowers post crypto collateral and choose leveraged positions based on the expected return on the underlying asset $\rho_U(\rho^*)$, risk aversion γ' , and the slope of the rate curve:

$$B = \frac{\rho_U(\rho^*) - f(u) - W'\gamma'\sigma_U^2}{\gamma'\sigma_U^2}.$$

In equilibrium, supply equals demand and utilization $u^* = B/S$ clears the market. The DeFi rate is then

$$\rho = u^* \cdot f(u^*) \cdot (1 - r), \tag{3}$$

where $f(u)$ is the protocol's base rate curve and r is the reserve factor.

The model delivers two competing channels for passthrough. A *supply channel* (Barbon

et al.’s emphasis) arises because higher ρ^* makes Treasuries more attractive, so LPs require higher DeFi rates to keep funds in the pool, leading to $\partial\rho/\partial\rho^* > 0$. A *demand channel* arises because higher ρ^* can depress expected crypto returns $\rho_U(\rho^*)$, reduce borrowing demand, lower utilization, and thereby lower DeFi rates, implying $\partial\rho/\partial\rho^* < 0$. The sign of passthrough in the model therefore depends on which channel dominates.

3.2.2 SMM Estimation

Constrained by a static single period model, there is no close form moment condition to be derived from barbons model. Instead, I compare empirical moments $\hat{\mathbf{m}}_d$ to moments $\hat{\mathbf{m}}_s(\boldsymbol{\theta})$ computed from simulated data (McFadden 1989):

$$\hat{\boldsymbol{\theta}}^{\text{SMM}} = \arg \min_{\boldsymbol{\theta}} [\hat{\mathbf{m}}_d - \hat{\mathbf{m}}_s(\boldsymbol{\theta})]' W [\hat{\mathbf{m}}_d - \hat{\mathbf{m}}_s(\boldsymbol{\theta})]. \quad (4)$$

I target four moments: (M1) the passthrough coefficient from regressing DeFi rates on SOFR; (M2) the utilization elasticity with respect to crypto returns; (M3) the standard deviation of DeFi rate changes; and (M4) the mean spread of DeFi rates over SOFR. The parameter vector is

$$\boldsymbol{\theta} = (\gamma, \gamma', \theta\psi, \varepsilon_\rho),$$

where ε_ρ governs the sensitivity of crypto returns to the risk-free rate. I set W to the identity matrix since the model is just-identified.

3.2.3 SMM Identification

Let $m(\boldsymbol{\theta})$ denote the vector of model-implied moments and $\hat{\mathbf{m}}_d$ the corresponding data moments. Because I estimate four parameters using four moments, the model is just-identified in the sense that the number of moments equals the number of parameters:

$$\hat{\mathbf{m}}^d = (\text{passthrough}, \text{utilization elasticity}, \text{rate volatility}, \text{mean spread}).$$

Formal identification requires:

1. **Order condition.** The number of moments is at least as large as the number of parameters. Here $4 = 4$, so the order condition holds.
2. **Rank condition.** The Jacobian of the moment vector with respect to the parameters, $G(\boldsymbol{\theta}) = \partial m(\boldsymbol{\theta}) / \partial \boldsymbol{\theta}'$, must have full column rank in a neighborhood of the true parameter vector. Intuitively, each parameter must shift at least one moment in a way that is not spanned by the effects of the other parameters.
3. **Model correctness.** The structural model must approximate the data-generating process for the targeted moments. In particular, the static equilibrium representation must be a reasonable abstraction of the dynamic DeFi rate-setting environment.

In this application, I do not claim to recover deep structural parameters. The model is highly stylized, treats a dynamic market as static, and abstracts from institutional details such as protocol governance, cross-chain liquidity, and off-chain collateral management. Instead, I use SMM as a diagnostic tool: given my moment selection and simulation design, I ask whether there exists *any* parameter vector $\boldsymbol{\theta}$ that can reproduce the four empirical moments. The answer is largely no. The estimated Jacobian is nearly singular in some directions, and the SMM objective is relatively flat in γ , γ' , and $\theta\psi$. As a result, the corresponding parameter estimates are weakly identified and have large standard errors.

Standard errors are computed using the SMM sandwich formula,

$$\text{Var}(\hat{\boldsymbol{\theta}}_{\text{SMM}}) \approx \left(1 + \frac{1}{S}\right) (G'WG)^{-1},$$

where S is the number of simulation draws used to evaluate the model moments. The $(1 + 1/S)$ factor inflates the variance relative to standard GMM to account for simulation noise. Because the model is just-identified, Hansen's J -statistic does not provide a meaningful

overidentification test; instead, model performance is evaluated by inspecting the size and sign of moment mismatches.

4 Empirical Results

4.1 Reduced-Form Results

Table 2 reports the OLS estimates. Column (1) shows the baseline specification. The coefficient on SOFR is 0.189 (s.e. = 0.050), indicating that a 100 basis point increase in the risk-free rate is associated with a 19 basis point increase in DeFi deposit rates. The associated t -statistic rejects the null hypothesis $H_0 : \beta_1 = 0$ at the 5% level, confirming statistically meaningful passthrough. Liquidation risk has a positive and significant coefficient (1.148), and its t -statistic similarly rejects $H_0 : \text{LiquidationRisk} = 0$, consistent with the model's prediction that higher liquidation risk raises the compensation LPs demand.

Table 2: OLS Estimates of Risk-Free Rate Passthrough

	(1)	(2)
	Baseline	Dynamic
SOFR	0.189*** (0.050)	0.132 (1.810)
Lagged DeFi Rate		0.805** (0.049)
Liquidation Risk	1.148*** (0.276)	1.208*** (0.271)
BTC Price	9.91e-06* (2.24e-06)	2.21e-05 (1.20e-05)
Gas Price	0.009** (0.004)	0.008 (0.004)
VIX	-0.034 (0.021)	-0.035 (0.019)
T5YIE	0.276 (0.182)	0.031 (0.193)
Coin FE	Yes	Yes
Implied Long-Run Passthrough	0.189	0.677

Notes: Dependent variable is the DeFi deposit rate (%). Standard errors clustered by stablecoin in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Column (2) adds the lagged dependent variable. The coefficient on the lag is 0.805, and the large t -statistic decisively rejects the null $H_0: \phi = 0$, indicating substantial persistence

in DeFi rate dynamics: a one percentage point shock to the DeFi rate today still shows up as roughly 0.8 percentage points tomorrow. The contemporaneous passthrough coefficient falls to 0.132, but the implied long-run passthrough is $0.132/(1 - 0.805) = 0.677$.

Higher gas prices are associated with higher DeFi rates (coefficient 0.009 in the baseline), and the corresponding t -statistic rejects $H_0 : \text{GasPrice} = 0$ only at the 10% level. This is consistent with transaction costs slowing arbitrage.

4.2 Structural Estimation Results

Table 3 reports the SMM parameter estimates. The estimates reveal serious identification problems as none of these parameters are statistically distinguishable from zero.

Table 3: SMM Parameter Estimates

Parameter	Estimate	Std. Error	95% CI	p -value
LP Risk Aversion (γ)	0.117	0.801	$[-1.45, 1.69]$	0.884
Borrower Risk Aversion (γ')	0.038	0.258	$[-0.47, 0.54]$	0.883
Liquidation Wedge ($\theta\psi$)	0.010	0.111	$[-0.21, 0.23]$	0.927
Crypto-Fed Elasticity (ε_ρ)	-1.023	0.016	$[-1.05, -0.99]$	0.000

Notes: Standard errors computed via the SMM sandwich formula with the $(1 + 1/S)$ simulation adjustment.

The weak identification is symptomatic of model misspecification. When the model cannot match the data moments, the objective function is flat in some directions, and the optimizer cannot pin down the parameters.

4.3 Model Fit

Table 4 compares the data moments to the model moments at the estimated parameters. The model fails on three of four moments, and the minimum value of the SMM criterion is

$Q(\hat{\theta}) = 3.3$, which is far from zero for a just-identified model that fits well.

Table 4: Moment Comparison: Data versus Model

Moment	Data	Model	Difference	Fit
M1: Passthrough (β_1)	0.635	-0.008	-0.643	Fails (wrong sign)
M2: Utilization Elasticity	2.796	2.479	-0.317	Reasonable
M3: Rate Volatility	0.029	0.006	-0.024	Fails (80% too low)
M4: Mean Spread	0.013	0.033	+0.020	Fails (148% too high)

Notes: Data moments computed from sample. Model moments computed from simulations at $\hat{\theta}_{\text{SMM}}$.

The passthrough failure is the most informative. The data show a passthrough of 0.635; the model generates -0.008. This sign reversal indicates that the parameter combination chosen by SMM leans heavily on the demand channel, effectively suppressing the LP arbitrage channel that Barbon et al. emphasize.

4.4 Why Does the Model Fail?

The moment mismatches reveal two major weaknesses in the static equilibrium framework.

First, the static model assumes that LPs can instantaneously rebalance between DeFi and Treasuries. Under this assumption, the supply channel should dominate: when the risk-free rate rises, LPs require higher DeFi rates to keep funds in the pool, leading to positive passthrough. In the data, however, DeFi rates are highly persistent: the estimated coefficient on the lagged dependent variable is 0.805. This persistence suggests that LP supply adjusts sluggishly. In the short run, utilization and rates are instead moved primarily by shocks to borrowing demand (for example, changes in crypto speculation), while supply catches up only gradually. A static model with instantaneous rebalancing cannot capture these dynamics.

Second, the static model has difficulty jointly matching rate volatility and mean spreads. It underpredicts volatility by about 80 percent and overpredicts mean spreads by roughly 150 percent. Intuitively, generating the observed high-frequency volatility in a static setting requires large shocks to utilization or liquidation risk, but these same shocks tend to produce spreads that are too large on average. The SMM procedure therefore compromises by choosing parameters that fit utilization elasticity reasonably well while sacrificing the sign of passthrough and the level and volatility of spreads.

Taken together, these failures suggest that DeFi rates are driven by utilization dynamics in a market where:

1. Borrowing demand responds quickly to crypto-specific news and risk appetite.
2. LPs adjust portfolios only gradually, so the supply channel is slow-moving.
3. Frictions such as gas costs and regulatory uncertainty slow arbitrage with Treasuries.

A dynamic structural model with adjustment costs and heterogeneous LPs is likely needed to reconcile the supply and demand channels and match the full set of moments.

5 Limitations

It is useful to summarize the main limitations of this analysis.

First, the rigorous causal reduced-form estimates rely on the assumption that, conditional on observables, daily movements in SOFR are exogenous to DeFi rates. I cannot rule out omitted factors that jointly drive both markets. With only three stablecoins, cluster-robust inference is also fragile, and alternative clustering schemes yield somewhat different p -values.

Second, the data are drawn from a single protocol (Aave) on a single blockchain (Ethereum) and focus on three large USD-pegged stablecoins. The results need not generalize to other protocols, chains, or non-USD collateral. In particular, periods of extreme market stress,

protocol upgrades, and regulatory shocks may change the mapping from SOFR to DeFi rates in ways not captured here.

Third, the structural model is intentionally static and highly stylized. I treat a high-frequency dynamic market as if it were in a sequence of static equilibria and match unconditional moments rather than fully specified transition dynamics. This simplification likely contributes to the model’s failure to match rate volatility and mean spreads and limits any quantitative policy interpretation of the estimated parameters.

Finally, the SMM estimation is best viewed as a validation rather than a structural estimation in the sense of Reiss and Wolak (2007). The fact that the model cannot jointly match the four targeted moments is informative about missing mechanisms, but it does not pin down the magnitudes of risk aversion, liquidation premia, or arbitrage elasticities in the underlying economy.

6 Conclusion

This paper tests the Barbon et al. (2025) structural model of DeFi lending. The reduced-form evidence documents positive passthrough from U.S. risk-free rates to stablecoin lending rates. The structural model, however, cannot match the data. The model generates a negative passthrough coefficient when the data show strong positive passthrough and fails on rate volatility and mean spreads. These failures reflect a fundamental tension: the static model assumes instantaneous LP rebalancing and a dominant supply channel, whereas the data reveal sluggish supply adjustment and important demand and frictional channels.

These failures are informative. They point to specific mechanisms that any successful structural model of DeFi rate determination must incorporate. The static equilibrium framework of Barbon et al. (2025) provides useful intuition about the forces at play in DeFi lending markets, but it cannot serve as a quantitative model for policy analysis without dynamic extension.

References

- Aldasoro, I., G. Cornelli, M. F. Minesso, L. Gambacorta, and M. M. Habib. 2025. “Stablecoins, Money Market Funds and Monetary Policy.” *Economics Letters*, 247: 112203.
- Angrist, Joshua D., and Jörn-Steffen Pischke. 2009. *Mostly Harmless Econometrics: An Empiricist’s Companion*. Princeton: Princeton University Press.
- Barbon, Andrea, Jean Barthélemy, and Benoit Nguyen. 2025. “DeFi-ying the Fed? Monetary Policy Transmission to Stablecoin Rates.” Banque de France Working Paper.
- Benigno, Pierpaolo, and Carlo Rosa. 2023. “The Bitcoin-Macro Disconnect.” *Journal of Monetary Economics*, forthcoming.
- Karau, Soeren. 2021. “Monetary Policy and Bitcoin.” Deutsche Bundesbank Discussion Paper 41/2021.
- McFadden, Daniel. 1989. “A Method of Simulated Moments for Estimation of Discrete Response Models Without Numerical Integration.” *Econometrica*, 57(5): 995–1026.
- Reiss, Peter C., and Frank A. Wolak. 2007. “Structural Econometric Modeling: Rationales and Examples from Industrial Organization.” In *Handbook of Econometrics*, Vol. 6A, ed. James J. Heckman and Edward E. Leamer, 4277–4415. Amsterdam: Elsevier.
- Watsky, Cy, Jeffrey Allen, Hamzah Daud, Jochen Demuth, Daniel Little, Megan Rodden, and Amber Seira. 2024. “Primary and Secondary Markets for Stablecoins.” *FEDS Notes*. Washington: Board of Governors of the Federal Reserve System, February 23, 2024. <https://doi.org/10.17016/2380-7172.3447>.